MA2108

AY25/26 Sem 1

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Ordered Fields

D1.1 Field A field is a set F with addition and multiplication. $\mathbb{R}, \mathbb{Q}$ are fields while $\mathbb{N}, \mathbb{Z}$ are not.

Addition Axioms Closure of Addition, Commutativity, Associativity, Identity Element, Additive Inverse

Multiplication Axioms Closure of Multiplication, Commutativity, Associativity, Identity Element, Multiplicative Inverse, Distributive Law

P1.4

- $x+y=x+z \Rightarrow y=z$
- $x+y=x \Rightarrow y=0$
- $x+y=0 \Rightarrow y=-x$
- $-(-x)=x$
- $x \neq 0 \wedge xy=xz \Rightarrow y=z$
- $x \neq 0 \wedge xy=x \Rightarrow y=1$
- $x \neq 0 \wedge xy=1 \Rightarrow y=1/x$
- $x \neq 0 \Rightarrow 1/(1/x)=x$
- $0x=0$
- $x \neq 0 \wedge y \neq 0 \Rightarrow xy \neq 0$
- $(-x)y=-(xy)=x(-y)$
- $(-x)(-y)=xy$

D1.5 Total Order A relation satisfying trichotomy and transitivity property. An ordered set is a set with a total order.

D1.8 Ordered Field A field where $y<z \Rightarrow x+y<x+z$, $x>0, y>0 \Rightarrow xy>0$. All ordered fields contain (a homomorphism of) $\mathbb{Q}$.

P1.12 Ordered Field Statements

- $x>0 \Rightarrow -x<0$ and vice versa
- $x>0 \wedge y<z \Rightarrow xy<xz$
- $x<0 \wedge y<z \Rightarrow xy>xz$
- $xy>0 \Rightarrow (x>0 \wedge y>0) \vee (x<0 \wedge y<0)$
- $x \neq 0 \Rightarrow x^2>0$
- $0<x<y \Rightarrow 0<1/y<1/x$

P1.15 There exists $\iota : \mathbb{Q} \hookrightarrow F$ such that ι is order preserving for all ordered fields.

Least Upper Bound Property

D1.16 Upper Bound Suppose S is an ordered set and $E \subset S$. If $\exists \beta \in S, \forall x \in E, x \leq \beta$, then E is bounded above and β is an upper bound of E. Lower bound is defined similarly with $\geq$. They might not be unique and could possibly not exist. They might not be in the set itself.

D1.19 Least Upper Bound Suppose $\alpha \in S$, α is an upper bound of E, and if $\gamma<\alpha$ then γ is not an upper bound of E. Then α is the least upper bound or supremum of E, and $\alpha = sup\ E$. We can similarly define the greatest lower bound or infimum $\alpha = inf\ E$. The LUB is also unique.

R1.20 Maximum α is a maximum of E if it is a LUB and $\alpha \in E$.

T1.26 Least Upper Bound Property An ordered set S where $\forall E \subseteq S, E \neq \emptyset$, and E is bounded above implies $\exists sup\ E \in S$. $\mathbb{Q}$ does not have this property.

T1.29 Completeness of $\mathbb{R}$ $\mathbb{R}$ is the ordered field which has the LUB property and is unique up to isomorphism. It also contains $\mathbb{Q}$ as a subfield.

T1.32 An ordered set with the LUB property also has the GLB property.

P1.34 Let $E \subset \mathbb{R}, a \in \mathbb{R}, T=\{a+x : x \in E\}$. If E is bounded above, then $sup\ T = a + sup\ E$.

- $\emptyset \neq E \subseteq \mathbb{R}$, E is bounded above, so $sup\ E$ exists
- $\forall x \in E, x \leq sup\ E, a+x \leq a + sup\ E$
- $a + sup\ E$ must be an upper bound for T, let M be any upper bound of T
- $\forall x \in E, a+x \leq M, x \leq M-a$
- $M-a$ is an upper bound for E
- By definition of supremum, $sup\ E \leq M-a, a+sup\ E \leq M, sup\ T \geq a + sup\ E$

R1.35 To show equality, either show $A \leq B \wedge B \leq A$ or $A \leq B$, $A<B$ is a contradiction

Archimedean Property

D2.1 Archimedean Property Let F be an ordered field. F has the Archimedean Property if $\forall a, b \in F, a>0, \exists n \in \mathbb{N}, na>b \in F$

D2.1 Density For an ordered field F, $\mathbb{Q}$ is dense in F wrt ordering if $\forall x, y \in F, x<y, \exists r \in \mathbb{Q}, x<r<y \in F$

P2.4 F has the Archimedean Property iff $\mathbb{Q}$ is dense in F

$\mathbb{Q}, \mathbb{R}$**:** $\mathbb{Q}$ is Archimedean and is dense in itself, while $\mathbb{R}$ is Archimedean as well

C2.6 $\forall \varepsilon \in \mathbb{R}_{>0}, \exists\, n \in \mathbb{N}, 0<\frac{1}{n}<\varepsilon$

T2.7 nth roots $\forall x \in \mathbb{R}_{>0}, n>0 \in \mathbb{Z}, \exists !\, y, y^n = x$

C2.8 For positive real a, b and positive integer n, $(ab)^{1/n} = a^{1/n}b^{1/n}$

C2.9 If $a, b \in \mathbb{R}, a<b, \exists\, x \in \mathbb{R} \backslash \mathbb{Q}, a<x<b$

P2.10 Let $\alpha \in \mathbb{R}, E=\{x \in \mathbb{Q} : x<\alpha\} \subseteq \mathbb{Q}$. Then $sup\ E = \alpha$

Sequences

P2.13 Properties of absolute value

- $|a| \geq 0, a \leq |a|, -a \leq |a|$
- $|a|=0 \Leftrightarrow a=0$
- $|-a|=|a|$
- $|ab|=|a| \cdot |b|$
- $|a|^2=a^2$
- $c \geq 0 \Rightarrow |a| \leq c \Leftrightarrow -c \leq a \leq c$
- $-|a| \leq a \leq |a|$

P2.14 Triangle Inequality $\forall a, b \in F, |a+b| \leq |a|+|b|$

C2.15 $\forall a, b \in F$,

- $||a|-|b|| \leq |a-b|$
- $|a-b| \leq |a|+|b|$
- $||a|-|b|| \leq |a+b|$

D2.16 Sequences A sequence in a set X is a function $x : \mathbb{N} \to X$, denoted by $(x_n)_n$ or $\{x_n\}_n$.

D2.21 Convergence A sequence in F converges iff $\exists x \in F, \forall \varepsilon \in F_{>0}, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq N \Rightarrow |x_n - x| < \varepsilon$. For every small distance ε, there is a cut-off N such that for all larger n, the absolute difference is smaller than ε. x is the limit of the sequence in F, and $lim_{n\to\infty} x_n = x$.

P2.23 The limit of a sequence is unique.

D2.24 Bounds A sequence is eventually constant if $\exists N \in \mathbb{N}, \forall n \geq N, x_n = x_N = x$. A sequence is bounded in F if the set of values is bounded in F by some B. We can define one sided bounds similarly.

P2.26 If a sequence is convergent in F, then it is bounded.

E2.29 Example Convergence Proof (Make n subject of formula)

$$\begin{aligned} \left|\frac{2n^2+1}{n^2+3n} - 2\right| &= \left|\frac{2n^2+1-2(n^2+3n)}{n^2+3n}\right| \\ &= \left|\frac{1-6n}{n^2+3n}\right| \\ &\leq \frac{1+6n}{n^2+3n} \\ &\leq \frac{n+6n}{n^2} = \frac{7n}{n^2} = \frac{7}{n} \end{aligned}$$

- Let $\varepsilon>0$ be given, then choose $K \in \mathbb{N}$ such that $K>\frac{7}{\varepsilon}$. Then $\forall n \geq K$,

$$\begin{aligned} \left|\frac{2n^2+1}{n^2+3n} - 2\right| &\leq \frac{7}{n} \text{ since } n \geq K \\ &\leq \frac{7}{K} \\ &< \varepsilon \text{ since } K > \frac{7}{\varepsilon} \end{aligned}$$

Convergent Sequences

P3.1 Properties of Limits of Convergent Sequences Let $(s_n)_n, (t_n)_n$ be convergent sequences in F. Then $(s_n+t_n)_n, (cs_n)_n, (s_nt_n)_n, (|s_n|)_n, (\sqrt{s_n})_n, (\frac{1}{s_n})_n$ are all convergent with limits following the substitution. If $\exists N \in \mathbb{N}, \forall n \geq N, s_n \leq t_n$, then $lim_{n\to\infty} s_n \leq lim_{n\to\infty} t_n$.

P3.2 Squeeze Theorem If $(x_n)_n, (y_n)_n, (z_n)_n$ are sequences in an ordered field F such that $\forall n, x_n \leq y_n \leq z_n$ and $lim_{n\to\infty} x_n = lim_{n\to\infty} z_n = a$, then $lim_{n\to\infty} y_n = a$.

More Sequences

E3.3 For real $0 \leq b < 1, lim_{n\to\infty} b^n = 0$.

E3.6 For real $c>0, lim_{n\to\infty} c^{\frac{1}{n}} = 1$.

E3.7 For real $a>1$ and integer $k>0$, $lim_{n\to\infty} \frac{n^k}{a^n} = 0$.

E3.8 $lim_{n\to\infty} n^{\frac{1}{n}} = 1$

D3.9 Monotone Sequences (s_n) is monotonically increasing if $\forall n, s_n \leq s_{n+1}$, while it is monotonically decreasing if $\forall n, s_n \geq s_{n+1}$.

T3.10 Monotone Convergence Every monotone sequence in $\mathbb{R}$ is bounded iff it converges.

Subsequences, Cauchy Sequences

D4.2 Subsequence Given a sequence in F, consider $(n_k)_k$ of positive integers where $n_1 < n_2 < ... < n_k < ....$ Then $(x_{n_k})_k$ is a subsequence of $(x_n)_n$.

E4.3 A sequence converges to x iff every subsequence also converges to x.

R4.5 For divergence, we can show a subsequence which diverges, or two subsequences which converge to different limits.

T4.9 Existence of Monotone Subsequence Let there be a sequence in an ordered field F which is not necessarily convergent. There exists a monotone subsequence in it.

T4.10 Bolzano-Weierstrass Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

D4.11 Cauchy Sequence A sequence $(p_n)_n$ in F is said to be Cauchy if $\forall \varepsilon \in F_{>0}, \exists N \in \mathbb{N}, \forall m, n \in \mathbb{N}_{\geq N}, |p_n - p_m| < \varepsilon$.

P4.13 If a sequence is convergent, then it is Cauchy.

P4.15 If a sequence is Cauchy, then it is bounded.

P4.16 Properties of Cauchy Sequences If $(s_n), (t_n)$ are Cauchy sequences, then $(s_n+t_n), (cs_n), (s_nt_n), (|s_n|)$ are also Cauchy.

More Cauchy Sequences

D4.18 Cauchy Complete An ordered field in F is Cauchy complete iff every Cauchy sequence in F is convergent in F. $\mathbb{R}$ is Cauchy complete.

P4.19 Let p_n be a Cauchy sequence in F. If there exists a subsequence of p_n that converges in F, then p_n also converges in F to the same limit.

T4.20 $\mathbb{R}$ is Cauchy complete.

D4.22 Contractive A sequence x_n is contractive if there exists C with $0<C<1$ such that $|x_{n+2} - x_{n+1}| \leq C|x_{n+1} - x_n|$ where C does not depend on n.

P4.23 Every contractive sequence is Cauchy and thus convergent.

D4.24 Extended Real Numbers The extended real number system consists of $\mathbb{R}$, $+\infty$, and $-\infty$. The original order is preserved and $\forall x \in \mathbb{R}, -\infty < x < +\infty$. This ordered set (not field) is denoted by $[-\infty, +\infty]$. The supremum and infimum of any subset always exists.

D4.28 Limit to Infinity The limit of x_n is $+\infty$ if for all real $M, \exists N, \forall n \geq N, s_n \geq M$. Similarly for $-\infty$.
P4.30 Let x_n be a sequence in $\mathbb{R}$, $x \in [-\infty, +\infty]$. Then $x_n \to x$ iff $\forall A \in [-\infty, \infty], A < x, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq N \Rightarrow A < x_n$ and the opposite holds.
L4.31 For any sequence $x_n \in \mathbb{R}$, there exists a subsequence x_{n_i} which converges in $[-\infty, +\infty]$.

limsup, liminf

D5.1 limsup, liminf Let x_n be a sequence in $\mathbb{R}$. Let $E = \{\text{the limits in } [-\infty, +\infty] \text{ of all convergent subsequences of } x_n\}$. Then define $\limsup_{n\to\infty} = sup(E), \liminf_{n\to\infty} = inf(E)$. $\limsup$ is the limit superior, while $\liminf$ is the limit inferior.
P5.3 Equivalent Definition Let x_n be a bounded series in $\mathbb{R}$ and $M = \limsup x_n$. $M = \limsup x_n$ if $M \in \mathbb{R}$ satisfies the following: For each $\varepsilon > 0$ there are at most finitely many n such that $x_n \geq M + \varepsilon$. Alternatively, $\forall \varepsilon, \exists K \in \mathbb{N}, \forall n \geq K, x_n < M + \varepsilon$. Also, for each $\varepsilon > 0$, there are infinitely many n such that $x_n > M - \varepsilon$. The same holds for $\liminf$.
P5.5 Convergence with limsup, liminf Let s_n be a sequence in $\mathbb{R}$. Then $\liminf_{n\to\infty} s_n \leq \limsup_{n\to\infty} s_n \in [-\infty, +\infty]$. Equality only holds iff s_n converges.
P5.7 If $s_n \leq t_n, \forall n \geq N, \liminf_{n\to\infty} s_n \leq \liminf_{n\to\infty} t_n$ and $\limsup_{n\to\infty} s_n \leq \limsup_{n\to\infty} t_n$
P5.8 Equivalent Definition of limsup Let x_n be a bounded sequence. For each $n \in \mathbb{N}$, let $y_n = sup\{x_k : k \geq n\}$. Then $\limsup_{n\to\infty} x_n = lim_{n\to\infty} y_n$.
P5.9 Equivalent Definition of liminf Let x_n be a bounded sequence. For each $n \in \mathbb{N}$, let $z_n = inf\{x_k : k \geq n\}$. Then z_n is increasing and bounded and $\liminf x_n = lim_{n\to\infty} z_n$

Series

D5.10 Series Let F be an ordered field. Given a sequence $a_n \in F$, we can form an infinite series $\sum_{k=1}^{\infty} a_k = a_1 + a_2 + \ldots$. More precisely, define $S_k = \sum_{n=1}^{k} a_n$ (nth partial sum).
P5.13 Properties of Convergent Series If $\sum a_n, \sum b_n$ are convergent series and $c \in F$, then $\sum(a_n + b_n)$, and $\sum ca_n$ are convergent.
P5.14 Cauchy Criterion for Series Let a_n be a sequence in $\mathbb{R}$. Then $\sum a_n$ converges iff $\forall \varepsilon > 0, \exists N, \forall m, n, m \geq n \geq N, |\sum_{k=n}^{m} a_k| \leq \varepsilon$.
C5.15 If $\sum a_n$ converges, $lim_{n\to\infty} a_n = 0$. However, the converse is false.
C5.17 nth term test If a_n does not converge or converges to a nonzero, then $\sum a_n$ diverges.
D5.19 A series eventually has a property (*) if $\exists N \in \mathbb{N}, \forall k \geq N, a_k$ has (*).
P5.20 Let $\sum a_n$ be an eventually non-negative series where $a_n \in \mathbb{R}$. Then a_n converges iff S_n is bounded above.
E5.21 Geometric Series For $\sum_{k=1}^{\infty} ar^{k-1}$, $s_n = \frac{a(1-r^n)}{1-r}$ if $r \neq 1$.
E5.22 p-series If $p > 1$, the p-series $\sum \frac{1}{n^p}$ converges. When $p \leq 1$, the series diverges.
D5.23 Absolute Convergence The series $\sum a_n$ converges absolutely if $\sum |a_n|$ converges. If a series converges but not absolutely, it converges conditionally.
P5.24 Let a_n be a sequence in $\mathbb{R}$. If $\sum a_n$ converges absolutely, then a_n converges. However, the converse is false.

Convergence Tests

P6.1 Direct Comparison Test Let a_k, b_k be sequences in $\mathbb{R}$ such that $\sum a_k, \sum b_k$ are eventually non-negative. Suppose we eventually have that $0 \leq a_k \leq b_k$. If $\sum b_k$ converges, then $\sum a_k$ converges. If $\sum a_k$ diverges, then $\sum b_k$ diverges.
P6.4 Limit Comparison Test Let a_k, b_k be sequences in $\mathbb{R}$ such that $\sum a_n, \sum b_n$ are eventually positive, and suppose the limit $\rho = lim_{n\to\infty} \frac{a_n}{b_n}$ exists. If $\rho > 0$, they either both converge or both diverge. If $\rho = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
P6.5 Ratio Test Let a_n be a sequence in $\mathbb{R}$ such that $\sum a_n$ is eventually positive, and let $\rho = \limsup \frac{a_{n+1}}{a_n}$. If $\rho < 1$, $\sum a_n$ converges. If $\exists N \in \mathbb{N}, \forall n \geq N, \frac{a_{n+1}}{a_n} \geq 1$, then the series diverges.
P6.8 Root Test Let a_k be a sequence in $\mathbb{R}$ such that $\sum a_n$ is eventually non-negative, and let $\rho = \limsup a_n^{1/n}$. If $\rho < 1$, then $\sum a_n$ converges. If $\rho > 1$, then $\sum a_n$ diverges. If $\rho = 1$, we cannot make a conclusion.
L6.9 For any sequence c_n of positive numbers, $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n\to\infty} \sqrt[n]{c_n} \leq \limsup_{n\to\infty} \sqrt[n]{c_n} \leq \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$.
T6.11 Dirichlet's Test Let a_n, b_n be sequences in $\mathbb{R}$. Suppose the partial sums A_n form a bounded sequence, b_n is monotone decreasing, and $lim\ b_n = 0$. Then $\sum a_n b_n$ converges.
L6.12 Partial Summation Given a_n, b_n, indexed by $Z \geq 0$, define $A_{-1} = 0$ and $\forall n \geq 0, A_n = \sum_{k=0}^{n} a_k$. Then for any $0 \leq p \leq q$, we have $\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q B_q - A_{p-1} b_p$.
E6.13 $\frac{\cos(n)}{\sqrt{n}}$ is convergent using Dirichlet's and $\sum_{k=1}^{n} \cos k = \frac{\sin(n+\frac{1}{2}) - \sin\frac{1}{2}}{2\sin\frac{1}{2}}$. $\sum_{k=1}^{n} \sin k = \frac{\sin(\frac{n}{2})\sin(\frac{n+1}{2})}{\sin\frac{1}{2}}$.
C6.14 Alternating Series Test Let $\sum(-1)^{n+1} b_n$ or $\sum(-1)^n b_n$ be an alternating series with b_k as a sequence in $\mathbb{R}$. Suppose $b_n \geq 0$, b_n is monotone decreasing, and $lim_{n\to\infty} b_n = 0$. Then it is convergent.

Rearrangement of Series

D6.15 Rearrangement Let σ be a permutation of $\mathbb{N}$. Then the series $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is a rearrangement of $\sum_{n=1}^{\infty} a_n$.
T6.16 Rearrangement of Aboslutely Convergent Series Let a_n be a sequence in $\mathbb{R}$, and $\sum a_k$ be an absolutely convergent series. Then any rearrangement of $\sum a_k$ also converges and has the same sum as $\sum a_k$.
T6.17 Riemann Rearrangement Let a_k be a sequence in $\mathbb{R}$ such that $\sum a_n$ is conditionally convergent. Let $\alpha, \beta \in [-\infty, +\infty], \alpha \leq \beta$. Then there exists a rearrangement $\sum b_n$ of $\sum a_n$ such that the sequence of partial sums s_n' of b_n satisfies $\liminf s_n' = \alpha, \limsup s_n' = \beta$.

Metric Spaces

D7.1 Metric Space A set X whose elements are points, is a metric space if for any two points $p, q \in X$, there is an associated real number $d(p, q)$, the distance from p to q, such that it satisfies:

1. Positivity: $d(p, q) > 0$ for $p \neq q, d(p, p) = 0$
2. Symmetry: $d(p, q) = d(q, p)$
3. Triangle Inequality: $d(p, q) \leq d(p, r) + d(r, q), r \in X$

Any function $d : X \times X \to \mathbb{R}$ satisfying these is called a distance function or metric.
E7.2 Inner Product Space A vector space V over a field $F = \mathbb{R}, \mathbb{C}$ with inner product $\langle -, - \rangle : V \times V \to F$ such that

1. Conjugate Symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$. If $F = \mathbb{R}$, it is just symmetry.
2. Linearity in the First Argument: $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.
3. Positive-Definiteness: $x \neq 0, \langle x, x \rangle > 0$

We can define the norm $|| \cdot || : V \to \mathbb{R}$ by $||x|| = \sqrt{\langle x, x \rangle}$. Taking $X = V, d(x, y) = ||x - y||$ defines a metric space.
E7.3 Euclidean Space Define an inner product on $\mathbb{R}^k$ for any $k \in \mathbb{N}$ by $\langle (x_1, \ldots, x_k), (y_1, \ldots, y_k) \rangle_{\mathbb{R}^k} = \sum_k x_k y_k$. An Euclidean space is a pair $(V, \langle -, - \rangle)$ where V is a vector space and $\langle -, - \rangle$ is a norm which is isomorphic as an inner product space to $(\mathbb{R}^k, \langle -, - \rangle_{\mathbb{R}^k})$.
D7.8 ε-neighbourhood Let (X, d_X) be a metric space. For $\varepsilon > 0, x \in X$, the ε-neighbourhood of x is $N_\varepsilon(x) := \{y \in X : d_X(x, y) < \varepsilon\}$.

D7.9 Convergence in Metric Space Let X be a metric space, and $(p_n)_{n\in\mathbb{N}}$ be a sequence in X. (p_n) converges in X to a point $p \in X$ if $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, n \geq N \Rightarrow d(p_n, p) < \varepsilon$. Alternatively, $(d(p_n, p))_{n\in\mathbb{N}} \to 0$ in $\mathbb{R}_{\geq 0}$.

L7.10 Uniqueness of Limits in Metric Spaces The limit of of a sequence in a metric space, if it exists, is unique.

D7.11 A set $E \subseteq X$ is bounded if there exists $M \in \mathbb{R}, q \in X$ such that $\forall p \in E, d(p, q) < M$.

D7.12 A sequence (p_n) in X is eventually constant if $\exists N \in \mathbb{N}$ such that $\forall n \geq N$ we have $p_n = p$. It is bounded in X if $\{p_n\}$ is bounded in X.

P7.13 Let (p_n) be a sequence in a metric space. If (p_n) converges, then it is bounded.

P7.14 (p_n) converges in X iff it is eventually constant for the discrete metric.

L7.15 (p_n) converges in X iff every subsequence of (p_n) converges in X. The limit of (p_n) is equal to the limit of every one of its subsequences.

D7.16 Cauchy A sequence (p_n) in X is Cauchy if $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}, \forall m, n \in \mathbb{N}, m, n \geq N \Rightarrow d(p_n, p_m) < \varepsilon$. The sequence $(d(p_n))$ is Cauchy in $\mathbb{R}$.

P7.17 If (p_n) is convergent in X, then it is Cauchy.

P7.18 If (p_n) is Cauchy in X, it is bounded.

P7.19 If (p_n) is Cauchy in X and there exists a subsequence that converges in X, then (p_n) also converges in X to the same limit.

D7.20 A metric space X is Cauchy complete iff every Cauchy sequence in X is convergent in X.

Limits in Metric Spaces

P8.1 Convergence in Euclidean Spaces Let $V \simeq \mathbb{R}^k$ be an Euclidean space with ordered basis B of size k. For each $i \leq k$, let $\pi_i : V \to \mathbb{R}$ denote the projection onto the ith coordinate. Then any sequence (p_n) in V converges in V to $p \in V$ if for each i, the sequence $(\pi_i(p_n))$ converges in $\mathbb{R}$ to $\pi_i(p) \in \mathbb{R}$. The converse also holds.

P8.3 Any Euclidean space is Cauchy complete.

T8.4 Let V be an Euclidean space and (p_n) a bounded sequence. Then (p_n) contains a convergent subsequence.

D8.5 Let $E \subset X$. $p \in X$ is a limit point of E if there exists a sequence (p_n) such that $p \neq p_n \in E$ for all n and $lim_{n\to\infty} p_n = p$. $p \in E$ is not required.

D8.6 Let $E \subset X, f : E \to Y$, and p be a limit point of E. We write $f(x) \to q \in Y$ as $x \to p$, or $lim_{x\to p} f(x) = q$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $d_Y(f(x), q) < \varepsilon$ for all points $x \in E$ for which $0 < d_X(x, p) < \delta$. $p \in X$ but not necessarily E. We also could have $f(p) \neq lim_{x\to p} f(x)$.

P8.7 Sequential Criterion for Limit Suppose $E \subset X, f : E \to Y$, p is a limit point of E. Then $lim_{x\to p} f(x) = q$ iff $lim_{n\to\infty} f(p_n) = q$ for every sequence $\{p_n\}$ in E such that $p_n \neq p, lim_{n\to\infty} p_n = p$.

P8.8 Suppose $E \subset X, f : E \to Y, p$ is a limit point of E. If there exists $q, q' \in Y'$ such that $lim_{x\to p} f(x) = q, q'$, then $q = q'$.

P8.10 Let $E \subset X, f, g : E \to \mathbb{R}$, and p be a limit point of E. Suppose $lim_{x\to p} f(x) = L$ and $lim_{x\to p} g(x) = M, L, M \in \mathbb{R}$. Let $k \in \mathbb{R}$ be a fixed constant. Then the sum, difference, scaling, composition, and division of limits holds.

Continuity

D8.11 Continuity Suppose $(X, d_X), (Y, d_Y)$ are metric spaces, $E \subset X, p \in E, f : E \to Y$. Then f is continuous at $p \in E$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $d_Y(f(x), f(p)) < \varepsilon$ for all points $x \in E$ for which $d_X(x, p) < \delta$. If p is a limit point of E, then f is continuous at p iff $lim_{x \to p} f(x) = f(p)$. If f is continuous at every point of E, then f is said to be continuous on E.
E8.12 Isometry A map $f : X \to Y$ is an isometry iff $\forall x_1, x_2 \in X, d_X(x_1, x_2) = d_Y(f(x_1), f(x_2))$. It is injective and continuous.
P8.15 Sequential Criterion for Continuity Let $(X, d_X), (Y, d_Y)$ be metric spaces and $f : X \to Y$ be a map. f is continuous at $p \in X$ iff $lim_{n \to \infty} f(p_n) = f(p)$ for every sequence (p_n) in X such that $lim_{n \to \infty} p_n = p$.
P8.17 Composition of Continuous Suppose X, Y, Z are metric spaces, $E \subset X, f : E \to Y, g : f(E) \to Z, h(x) = g(f(x))$. If f is continuous at $p \in E$ and g is continuous at $f(p)$, then h is continuous at p.
P8.18 The function $(f_1, ..., f_k) : X \to Y^k$ defined by $p \to (f_1(p), ..., f_k(p))$ is continuous iff all f_i are continuous.
P8.20 The maps $+, -, \times, (-)^{-1}$ are all continuous.
C8.21 Let $f, g : X \to \mathbb{R}$ be continuous functions on X. Then $f + g, -f, fg, \frac{1}{f}$ are all continuous.
P8.22 Let X be a metric space, V be a Euclidean space, $f, g : X \to V$ be V-valued continuous functions on X, $\alpha : X \to \mathbb{R}$ be $\mathbb{R}$-valued continuous function on X. Then $f + g, -f, \alpha f, f \cdot g, ||f||$ are all continuous.
D8.23 Limits to $\pm\infty$ Let $E \subseteq X, f : E \to \mathbb{R}$ be a function and p be a limit point of E. $f(x)$ tends to $+\infty$ as $x \to p$ if for every $M > 0$, there exists $\delta > 0$ such that for all $x \in X$ and $0 < d(x, p) < \delta$ we have $f(x) > M$. Then $lim_{x \to p} f(x) = +\infty$.
P8.24 Let X be a metric space, $E \subseteq X, f : E \to \mathbb{R}$, p be a limit point of E. Then $lim_{x \to p} f(x) = +\infty$ iff for every sequence $\{p_n\}$ in $E \backslash \{p\}$ satisfying $lim_{n \to \infty} p_n = p$ one has $lim_{n \to \infty} f(p_n) = +\infty$.

Real-Valued Functions

D9.1/9.2 Let p be a limit point of $X \cap (p, \infty)$. L is the right-hand limit of f at p if for any $\varepsilon > 0$, there exists $\delta > 0$ such that $x \in X, p < x < p + \delta \Rightarrow |f(x) - L| < \varepsilon$. Then $lim_{x \to p^+} f(x) = L$. The left-hand limit has a similar definition.
D9.3 Let $\emptyset \subsetneq X \subset \mathbb{R}, f : X \to \mathbb{R}$ be a function. Suppose X is not bounded above. We say that L is the limit of f as $x \to \infty$ if for any given $\varepsilon > 0$, there exists $M > 0$ such that $x \in X, x > M \Rightarrow |f(x) - L| < \varepsilon, lim_{x \to \infty} f(x) = L$. Similarly for the not bounded below case with $x < M$.
D9.4 Let $f : X \to \mathbb{R}$ be a real-valued function. f is bounded iff the image $f(X)$ is a bounded set, i.e. there exists $M > 0$ such that $\forall x \in A, |f(x)| \leq M$. f is bounded on A if the image $f(A)$ is bounded.
P9.5 Let $f : [a, b] \to \mathbb{R}$ be a continuous function on the closed bounded interval $[a, b]$. Then f is bounded on $[a, b]$.
T9.6 Extreme Value Theorem Suppose $f : [a, b] \to \mathbb{R}$ is a continuous function on the closed bounded interval $[a, b]$. Then f has an absolute maximum and absolute minimum on $[a, b]$. f must be continuous, X must be bounded, closed, and $\mathbb{R}$ must be complete.
T9.8 Intermediate Value Theorem Suppose $f : [a, b] \to \mathbb{R}$ is continuous on the closed bounded interval $[a, b]$. Then for $f(a) < L < f(b)$ or the other way around, there exists $c \in (a, b)$ such that $f(c) = L$. f must be continuous, X must be connected, $\mathbb{R}$ must be complete.
P9.10 Let $a, b \in \mathbb{R}, a < b$. Suppose $f : [a, b] \to \mathbb{R}$ is continuous on the closed bounded interval $[a, b]$. Then the image $f([a, b])$ is also a closed bounded interval.
L9.11 An interval is a subset I of $\mathbb{R}$ such that $\forall x, y, \in I, x \leq y \Rightarrow [x, y] \subseteq I$.

P9.12 Preservation of Intervals Let I be an interval in $\mathbb{R}$, and suppose a function $f : I \to \mathbb{R}$ is continuous on I. Then $f(I)$ is an interval. It might not have the same form however.
P9.14 Continuous Inverse Theorem Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be a strictly monotone function. If f is continuous on I and $J = f(I)$, then its inverse function $f^{-1} : J \to \mathbb{R}$ is strictly monotone and continuous on J.
D9.16 Let X, Y be metric spaces. A function $f : X \to Y$ is uniformly continuous on X if for any given $\varepsilon > 0$, there exists $\delta > 0$ such that $x, u \in X, d_X(x, u) < \delta \Rightarrow d_Y(f(x), f(u)) < \varepsilon$. δ only depends on ε. Any uniformly continuous function is continuous.
P9.18 Sequential Criterion for Uniform Continuity Let $f : X \to Y$ be a function. TFAE: f is uniformly continuous on X; for any two sequences $(x_n), (u_n) \in X$ such that $d_X(x_n, u_n) \to 0$, we have $d_Y(f(x_n), f(u_n)) \to 0$.
C9.19 If $X \subseteq \mathbb{R}$, then any function $f : X \to \mathbb{R}$ is not uniformly continuous on X iff there exists sequences $(x_n), (u_n) \in X$ such that $x_n - u_n \to 0$ but $f(x_n) - f(u_n) \not\to 0$.
P9.21 Let $f : [a, b] \to \mathbb{R}$ be continuous on a closed bounded interval $[a, b]$. Then f is uniformly continuous on $[a, b]$.
P9.22 Let $f : X \to Y$ be uniformly continuous. If (x_n) is Cauchy in X, then $(f(x_n))$ is Cauchy in Y.
P9.23 A function $f : (a, b) \to \mathbb{R}$ is uniformly continuous iff the limits $L_a = lim_{x \to a} f(x)$ and $L_b = lim_{x \to b} f(x)$ exist and $\tilde{f} : [a, b] \to \mathbb{R}$ defined by joining $f(x), L_a, L_b$ is continuous.

Topology

D10.1 Topology A topology on a set X is a collection $\mathcal{T}$ of subsets of X with the following properties:

1. $\emptyset, X \in \mathcal{T}$.
2. The union of elements of any subcollection of T is in T.
3. The intersection of the elements of any finite subcollection of $\mathcal{T}$ is in $\mathcal{T}$.

A set X for which a topology $\mathcal{T}$ has been specified is a topological space specified by $(X, \mathcal{T})$. $U \subseteq X$ is an open set of X if U belongs to collection $\mathcal{T}$.
D10.2 Closed Subsets Let $(X, \mathcal{T})$ be a topological space. $Z \subseteq X$ is closed if the complement $X - Z$ is an open set.
E10.3 The discrete toplogy is the power set while the trivial topology is the set and the null set.
E10.6 Finite Complement Topology
$\mathcal{T}_f = \{U \subseteq X, X - U \text{ is finite or is all of } X\}$
D10.7 If $\mathcal{T} \subseteq \mathcal{T}'$, then $\mathcal{T}'$ is finer than $\mathcal{T}$ and T is coarser than $\mathcal{T}'$. The discrete toplogy is the finest possible topology.
E10.8 The metric topology is $\mathcal{T}_d = \{U \subseteq X : \forall p \in U, \exists r > 0, N_r(p) \subseteq U\}$
L10.9 $\forall p \in X, r > 0$, the ε-neighbourhood $N_r(p)$ is open in the metric topology.
P10.11 Let (X, d) be a metric space. $E \subseteq X$ is closed iff every limit point of E is contained in E.
P10.16 Union and Intersection of Closed Sets Let X be a topological space. Then the intersection of elements of any collection of closed sets or the union of any finite collection of closed sets is a closed set.
D10.17 Closure Let X be a topological space with $E \subseteq X$. The closure $\overline{E}$ of E is the intersection of all closed subsets of X which contain E. $E = \overline{E}$ iff E is closed.
D10.18 Interior Let X be a topological space with $E \subseteq X$. A point $p \in X$ is an interior point if there exists an open set U such that $p \in U, U \subseteq E$. The interior of E, $IntE$, is the union of all interior points of E.
X10.19 A subset E is open iff every point $x \in E$ is an interior point.

Basis

P11.1 Intersection of Topologies For any non-empty collcetion $Z = \{\mathcal{T}_i\}_{i \in I}$ of topologies on X, the intersection $\mathcal{T} = \bigcap_{i \in I} \mathcal{T}_i$ is also a topology on X.

D11.2 Let $S \subseteq \mathcal{P}(X)$ be any collection of subsets of X. Consider the collection J_S of all topologies $\mathcal{T}$ on X with S which is nonempty. Then $\mathcal{T}_S = \bigcap_{j \in J_S} \mathcal{T}_j$ is the topology generated by S, and is the coarsest topology on X where every member of S is open.

D11.5 Basis Let $\mathcal{B}$ be a collection of subsets of X such that:

1. For each $x \in X$, there is at least one $B \in \mathcal{B}$ such that $x \in B \Leftrightarrow X = \bigcup_{B \in \mathcal{B}} B$.
2. If $B_1, B_2 \in \mathcal{B}, x \in B_1 \cap B_2$, then there is $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Then we claim the topology $\mathcal{T}_{\mathcal{B}}$ generated by $\mathcal{B}$ is as follows: A subset U of X is open iff for each $x \in U$, there is $B \in \mathcal{B}$ such that $x \in B$ and $B \subseteq U$. $\mathcal{B}$ is a basis for the topology $\mathcal{T}_{\mathcal{B}}$.

C11.6 Let X be a set and $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on X. Then $\mathcal{T}$ is the collection of all unions of elements of $\mathcal{B}$.

E11.7 A basis for the metric topology of X is $\mathcal{B} = \{N_r(p) \in \mathcal{P}(X) : p \in X, r \in \mathbb{R}_{\geq 0}\}$, all open neighbourhoods in X.

Continuity and Homeomorphisms

D11.8 Topological Continuity Let X, Y be topological spaces. $f : X \to Y$ is continuous if for each open subset V of Y, $f^{-1}(V) = \{x \in X : f(x) \in V\}$ is an open subset of X.

P11.10 If $f : X \to Y, g : Y \to Z$ are continuous, then $g \circ f : X \to Z$ also is.

P11.12 Suppose X and Y are metric spaces with metrics d_X, d_Y. $f : X \to Y$ is continuous wrt the metric topologies iff $\forall p \in X, \forall \varepsilon > 0, \exists \delta > 0$ such that $\forall x \in X, d_X(x, p) < \delta$, we have $d_Y(f(x), f(p)) < \varepsilon$.

D11.13 Subspace Topology Let X be a topological space with topology $\mathcal{T}$. If A is a subset of X, $\mathcal{T}_A = \{A \cap U : U \in \mathcal{T}\}$ is the subspace topology on A.

P11.14 If A is a subspace of X, the inclusion function $j : A \to X$ is continuous. The subspace topology is the coarsest topology on X such that the inclusion map is continuous.

P11.16 If $f : X \to Y$ is continuous and if A is a subspace of X, then the restricted function $f|_A : A \to Y$ is continuous using $f \circ j$.

P11.17 Let $f : X \to Y$ be continuous. If Z is a topological space having Y as a subspace, then $h : X \to Z$ obtained by expanding the range of f is continuous using $h = i \circ f$.

P11.18 Let X, Y be topological spaces, and $A \subset Y$ be a subset with subspace topology. For any map $f_0 : X \to A$, TFAE: The map is continuous, or the composition $i \circ f_0 : X \to Y$ is continuous where $i : A \to Y$ is the inclusion map.

D11.19 Homeomorphism A continuous map $f : X \to Y$ is a homeomorphism iff there exists a continuous map $g : Y \to X$ such that $g \circ f = id_X$ and $f \circ g = id_Y$. It must be bijective.

D11.21 Homeomorphic Topological Subspaces X, Y are homeomorphic iff there exists a homeomorphism $f : X \to Y$. Properties of topological spaces invariant under homeomorphisms are called topological properties.

Connectedness

D12.1 A separation of X is a pair U, V of disjoin nonempty open subsets of X whose union is X. X is said to be connected iff there does not exist a separation of X.
D12.3 $A, B \subseteq X$ are separated if both $A \cap \overline{B}, \overline{A} \cap B$ are empty.
P12.4 $E \subset X$ is connected if E is not a union of two nonempty separated sets.
E12.5 If the topology of X is trivial, then it is connected. If X is empty or a singleton, it is connected. If X is discrete, it is connected iff it is empty or a singleton.
E12.6 The connected subsets of $\mathbb{R}$ are exactly the intervals, so $\mathbb{R}$ is connected.
R12.7 A subspace of a connected space need not be connected.
T12.8 Intermediate Value Theorem Let X, Y be topological spaces. If X is connected and $f : X \to Y$ is a continuous map, then $f(X)$ is connected.
C12.10 If X, Y are homeomorphic, X is connected iff Y is connected.
D12.12 Path Connected A path in X from x to y is a continuous map $f : [a, b] \to X$ of some closed interval in the real line into X such that $f(a) = x, f(b) = y$. A space X is path connected if every pair of points of X can be joined by a path in X.
P12.13 Every path connected space X is connected. The converse is not true.
D12.16 For $x, y \in \mathbb{R}^k$, define $[x, y] := \{\lambda x + (1-\lambda)y : \lambda \in [0, 1]\}$. It is connected. $E \subset \mathbb{R}^k$ is convex if $\forall x, y \in E, [x, y] \subset E$.
P12.18 Convex subsets of $\mathbb{R}^k$ are connected.

Hausdorff Spaces, Compactness

D12.19 Open Neighbourhood Let $x \in X$. An open subset $U \subseteq X$ which contains x is an open neighbourhood of x.
D12.20 Convergence in Topological Spaces Let (x_n) be a sequence in X and $x \in X$. (x_n) converges to $x \in X$ if for every open neighbourhood U of x, there exists $N \in \mathbb{N}$ such that for all $n \geq N, x_n \in U$.
P12.21 If X is a Hausdorff space, then any sequence (x_n) in X converges to at most one point.
D12.24 Hausdorff X is a Hausdorff space if for each pair $(x_1, x_2) \in X, x_1 \neq x_2$, there exists open neighbourhoods U_1, U_2 of x_1, x_2 respectively such that $U_1 \cap U_2 = \emptyset$.
E12.26 If X is a metric space, then X is Hausdorff.
P12.27 A subspace of a Hausdorff space is a Hausdorff space.
P12.28 The property of being Hausdorff is a topological property.
D12.29 A collection $\mathcal{A}$ of subsets of X covers X iff the union of the elements of $\mathcal{A}$ is equal to X. It is an open covering if $U \in \mathcal{A}$ is open.
D12.30 Compactness X is compact iff every open covering $\mathcal{A}$ of X contains a finite subcollection that also covers X.
E12.31 If the topology of X is trival then X is compact.
E12.32 If the topology of X is discrete then X is compact iff X is finite.
E12.33 Any finite topological space X is compact.
E12.34 $\mathbb{R}$ with the standard topology is not compact.
E12.35 $[a, b]$ is compact.
E12.36 $X = \{0\} \cup \{\frac{1}{n} : n \in \mathbb{N}\}$ is compact.
D12.37 $Y \subseteq X$ is compact if Y with the subspace topology is compact. A collection $\mathcal{A}$ of subsets of X covers Y if $\bigcup_{U \in A} U \supseteq Y$, and Y is compact if every open cover by sets in X has a finite subcover.
R12.38 The subspace of a compact space need not be compact.
P12.39 A finite union of compact subsets is compact.
P12.40 Every closed subspace of a compact space is compact.
P12.41 Every compact subspace of a Hausdorff space is closed.
L12.42 If Y is a compact subspace of a Hausdorff space X and $x_0 \notin Y$, then there exists disjoint open sets U, V with $x_0 \in U, Y \subseteq V$.
E12.43 For disjoint compact sets A, B in Hausdorff X, there exists disjoint open $U \supseteq A$ and $V \supseteq B$.
E12.44 In $\mathbb{R}$, only the closed interval $[a, b]$ is compact.
E12.46 In finite-complement topology, all subsets are compact but only finite sets are closed. This is not Hausdorff if X is not finite.
D12.47 Sequential Compactness A topological space X is said to be sequentially compact if every sequence in X has a convergent subsequence whose limit is X. $[a, b]$ satisfies this.
E12.48 The closed bounded interval $[a, b]$ is sequentially compact.
T12.49 Let X be a metric space. Suppose $A \subseteq X$ is sequentially compact. Then A is closed and bounded in X.

Compact subsets of $\mathbb{R}^n$

T13.1 Heine-Borel Theorem A subspace A of $\mathbb{R}^n$ is compact iff it is closed and bounded.
P13.2 If X is a metric space, then X is compact iff it is sequentially compact.
E13.3 The unit sphere S^{n-1} and closed unit ball B^n in $\mathbb{R}^n$ are compact because they are closed and bounded.
T13.4 Extreme Value Theorem Let X be any compact topological space and $f : X \to Y$ be continuous. Then the image $f(X)$ is compact.
R13.5 When $Y = \mathbb{R}, X = [a, b]$, since X is compact, $f(X)$ is compact and hence closed and bounded. So $\alpha = \sup f(X)$ exists, and since $f(X)$ is closed, $\alpha \in f(X)$, so there exists some $c \in [a, b]$ such that $f(c) = \alpha$. This gives the maximum.
C13.6 Compactness is a topological property.

Random Tutorial Crap

Bernoulli's Inequality $\forall x \geq -1, \forall n \in \mathbb{N}, (1+x)^n \geq 1 + nx$
T1P7 $sup\ X = -inf\ -X$
T2P4 $lim_{n\to\infty}|x_n| = 0$ iff $lim_{n\to\infty} x_n = 0$.
T3P5 x_n converges to $x \in \mathbb{R}$ iff x_{n_k} converges to x for every subsequence of x_n.
T3P6 If every subsequence of x_n has a subsequence which converges to 0, then x_n converges and $lim_{n\to\infty} x_n = 0$.
T4P1 $lim_n\ cos\ n$ does not exist.
T4P6 $\limsup(x_n + y_n) \leq \limsup x_n + \limsup y_n$, but equality might not hold.
T7P3 $\varepsilon - \delta$ Let $\varepsilon > 0$ be given.

$$\left|\frac{(2x+1)(x-2)}{(6x+2)} - (-1)\right| = \left|\frac{2x^2 - 3x - 2 + 6x + 2}{6x+2}\right|$$
$$= \frac{|x||2x+3|}{2|3x+1|}$$

Restrict x to $0 < |x - 0| < \frac{1}{4}$. Then $\frac{5}{2} < 2x + 3 < \frac{7}{2}, \frac{1}{4} < 3x + 1 < \frac{7}{4}$. $|2x+3| < \frac{7}{2}, |3x+1| > \frac{1}{4}$. Then $|\frac{(2x+1)(x-2)}{(6x+2)} - (-1)| < \frac{7/2}{2 \cdot 1/4}|x| = 7|x - 0|$. Now set $\delta = \min(\frac{1}{4}, \frac{\varepsilon}{7}) > 0$. Then $|\frac{(2x+1)(x-2)}{(6x+2)} - (-1)| < 7 \times \frac{\varepsilon}{7} = \varepsilon$. For numerator, pick upper bound, for denominator, pick lower bound.
T7P5 Let $f : A \to \mathbb{R}$ be a function and a be a limit point of A. If $\lim_{x\to a} |f(x)| = 0$ then $\lim_{x \to a} f(x) = 0$
T8P3 $h(x) = (x - \lfloor x \rfloor) \sin(\pi x)$ is continuous although $(x - \lfloor x \rfloor)$ is not.
T8P6 Suppose $h(x) = \frac{1}{q}, x \in \mathbb{Q} \cap (0, +\infty)$ and x is rational. We can use it to obtain (x_n) of irrational positive numbers that converge to $a \in \mathbb{Q} \cap (0, +\infty)$.
T9P5 Show there exists a point where $f(c) = f(c+3)$ given $f(0) = f(6)$. Let $g(x) = f(x) - f(x+3)$. Then $g(0) = f(0) - f(3), g(3) = f(3) - f(6) = f(3) - f(0) = -g(0)$. 0 must be a value between $g(0), g(3)$. Then by IVT, there exists c such that $g(c) = f(c) - f(c+3) = 0$.
T9P7 Suppose g is continuous and injective on $[0, 5]$, and $g(0) > g(5)$. Show g is strictly decreasing on $[0, 5]$. Claim $g(0) > g(x) > g(5), 0 < x < 5$. Since g is injective, $g(x) \neq g(0), g(5)$. Suppose that does not hold, then either $g(x) > g(0)$ or $g(x) < g(5)$. If $g(x) > g(0) > g(5)$, by IVT, there exists $x' \in (x, 5), g(x') = g(0)$, contradiction. If $g(0) > g(5) > g(x)$, by IVT, there exists $x' \in (0, x), g(x') = g(5)$, contradiction. To prove g is strictly decreasing, take $x, u \in [0, 5], x < u$. Suppose $g(x) < g(u)$. Since $g(0) > g(u) > g(x)$, by IVT, there exists $x' \in (0, x), g(x') = g(u)$, contradiciton. So $g(x) \geq g(u)$, and since $g(x) \neq g(u)$, g must be strictly decreasing.
T10P4 The Lipschitz condition says that $\exists K > 0, |f(x) - f(y)| \leq K|x - y|, \forall x, y \in A$. Choose $\delta = \frac{\varepsilon}{K} > 0$. Then $x, u \in A, |x - u| < \delta \Rightarrow |f(x) - f(y)| \leq K|x - y| < K \cdot \delta = K \cdot \frac{\varepsilon}{K} = \varepsilon$, and f is uniformly continuous on A. Find the choice of δ based on ε and distance.
T11P1 $\mathbb{R} - \mathbb{N} = \bigcup_{z \in \mathbb{N}}(z - 1, z) \bigcup (-\infty, 0)$ so $\mathbb{R} - \mathbb{N}$ is open and $\mathbb{N}$ is closed.
T11P2 Let $f : X \to \mathbb{R}$ be a continuous function and $A \subset X$ be open. If $f(A)$ is a single point, then $f(\overline{A})$ is also a single point.
T11P3 If A, B are disjoint compact subspaces of the Hausdorff space X, then there exists disjoin open sets U, V containing A, B respectively.
T11P5 Let $\{A_n\}_{n\in\mathbb{N}}$ be a sequence of connected subspaces of X, such that $A_n \bigcap A_{n+1} \neq \emptyset$. Then $\bigcup A_n$ is connected.
T11P6 No pair of $(0, 1), (0, 1], [0, 1]$ are homeomorphic. Only $[0, 1]$ is compact. $(0, 1) - \{g(1)\}$ is not connected.
T11P7 Let $f : X \to \mathbb{R}$ be a continuous map where X is the unit circle outline. Then there exists a point $(x, y) \in X, f(x, y) = f(-x, -y)$.
Extra P1 $\mathbb{Q}$ is neither open or closed in $\mathbb{R}$. $(q - \varepsilon, q + \varepsilon)$ always contains irrationals so no point $q \in \mathbb{Q}$ has a small open interval around it that sits entirely in $\mathbb{Q}$, so $\mathbb{Q}$ is not open. A set is closed if it contains all of its limit points. There exists a sequence of rationals that tends to $\sqrt{2} \notin \mathbb{Q}$, so $\mathbb{Q}$ is not closed.
Extra P2 $\mathbb{Q}$ satisfies $Int(\mathbb{Q}) = \emptyset, \overline{\mathbb{Q}} = \mathbb{R}$.
Extra P3 If $F \subset \mathbb{R}$ is closed, nonempty and bounded above, then $\sup F \in F$.
Extra P6 A finite union of compact subspaces of X is compact.
Extra P8 The intersection of an arbitrary collection of compact sets in $\mathbb{R}^k$ is compact.
Extra P10 If $f : X \to Y$ is continuous, where X is compact Y is Hausdorff, then f is a closed map (from a closed set to another closed set).
Review 1 If $f : X \to Y$ is continuous, then for every open $U \subseteq X$, $f(U)$ is open in Y is not necessarily true.
Review 2 $f(x) = \sqrt{x - 1}$ is uniformly continuous on $[1, \infty)$. We have $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$. $|f(x) - f(y)| = |\sqrt{x-1} - \sqrt{y-1}| \leq \sqrt{|(x-1) - (y-1)|} = \sqrt{|x - y|} < \sqrt{\delta} = \varepsilon$ for $\delta = \varepsilon^2$.